

Faithful on Paper, Unobservable in Practice: Türkiye's Transposition of Directive 2010/63/EU on Animals Used for Scientific Purposes

Summary: 1. Introduction. 2. The legal framework and its scope. 3. Authorisation and project evaluation. 4. Severity classification and the 3Rs. 5. Personnel competence. 6. Inspection, sanctions and reporting. 7. Assessment: six divergences and an observability gap. 8. Conclusions. Bibliography.

1. Introduction

Türkiye has never been bound by Directive 2010/63/EU on the protection of animals used for scientific purposes, yet its law on the subject was written to match it.¹ The Directive falls within accession Chapter 12 (food safety, veterinary and phytosanitary policy), which was screened in 2006–2007 and opened for negotiation in 2010.² Accession negotiations have been effectively stalled since 2018. Alignment with the Directive has nevertheless persisted, and it has been sustained less by accession leverage than by the expectations of international research sponsors, accreditation bodies and regulators who must decide whether data generated in a Turkish facility can be accepted without qualification.

The question is therefore not merely academic. Multinational sponsors that conduct or outsource preclinical work expect data that are defensible under the standards their own regulators apply, wherever the work was performed, and a Turkish facility working with such sponsors typically absorbs equivalent expectations by contract rather than because domestic law requires them. The same question determines whether a sponsor can rely on local ethics approval in place of its own audit, and whether the training of a Turkish researcher is recognised across borders. International benchmarks reinforce this asymmetry only unevenly. Accreditation by AAALAC International exists in Türkiye but is limited to a single institution, and no systematic pathway links Turkish competence certificates to the function categories used by the Federation of European Laboratory Animal Science Associations (FELASA). Delivery of FELASA-type training in Türkiye has instead been occasional, as in a 2013 joint course between Utrecht University and Dokuz Eylül University.³ In practice, compliance is shaped less by statute than by which sponsors, accreditors and export markets a given facility answers to. Whether the statute itself would suffice, if applied, is the question this article addresses.

This article asks how closely Turkish law on animals used for scientific purposes follows Directive 2010/63/EU, where it departs from it, and whether its performance can be observed from outside. It compares the Directive, article by article, with four Turkish instruments: the Animal Protection Law ("Hayvanları Koruma Kanunu", Law No. 5199 of 2004); the 2011 regulation on the welfare and protection of animals used for experimental and other scientific purposes (hereinafter the Welfare Regulation); the 2014 regulation on the working procedures and principles of animal-experiment ethics committees (hereinafter the Ethics Committee Regulation), which governs the central committee (HADMEK) and the local committees (HADYEK); and the 2019 regulation on aquatic vertebrates used for scientific purposes (hereinafter the Aquatic Regulation).⁴

¹European Union, Directive 2010/63/EU of the European Parliament and of the Council of 22 September 2010 on the protection of animals used for scientific purposes, *Official Journal of the European Union* L 276, 20 October 2010, 33–79 (hereinafter Directive 2010/63/EU).

²European Commission, Directorate-General for Neighbourhood and Enlargement Negotiations, *Screening Report Turkey: Chapter 12 – Food Safety, Veterinary and Phytosanitary Policy* (Brussels: European Commission, 2007), https://enlargement.ec.europa.eu/screening-report-turkey-chapter-12-food-safety-veterinary-and-phytosanitary-policy_en.

³AAALAC International, "Directory of Accredited Organizations," accessed 26 August 2026, <https://www.aaalac.org/accreditation/directory/directory-of-accredited-organizations-search-result/>; FELASA, "Education & Training," accessed 26 August 2026, <https://felasa.eu/education-training>.

⁴Türkiye, Hayvanları Koruma Kanunu [Animal Protection Law], Law No. 5199 of 24 June 2004, *Resmî Gazete* No. 25509, 1 July 2004, consolidated text, accessed 28 September 2026, <https://www.mevzuat.gov.tr/mevzuat?>

The method is doctrinal. Each Turkish provision was read in its original text in the official gazette ("Resmî Gazete") and then checked against the consolidated text in force in September 2026. The Welfare Regulation was amended in 2012 and 2019, the Aquatic Regulation in 2022 and 2024, and the Ethics Committee Regulation has not been amended.⁵ The legal analysis is supplemented by the activity reports and statistics of the central ethics committee, by three local committee rulebooks, and by the secondary literature.

The closest earlier comparison, by Gölcü and Aksoy, concluded from the texts alone that parts of both Turkish regulations were translated almost word for word from the Directive, and that without adaptation to Turkish conditions they might not provide an adequate basis of protection.⁶ This article adds a provision-level comparison, a full reading of Law No. 5199, the consolidated texts and the available enforcement data. Its argument is that the Turkish framework is largely adequate on paper, that six narrower divergences survive, and that the most consequential gap is not textual but observational: whether the rules are applied cannot be verified from public sources.

2. The legal framework and its scope

2.1. The Directive

Directive 2010/63/EU applies where animals are used or intended to be used in procedures, or bred specifically so that their organs or tissues may be used for scientific purposes.⁷ It covers live non-human vertebrates, including independently feeding larval forms and foetal forms of mammals from the last third of their normal development, and live cephalopods.⁸ Its scope is defined by the procedure as much as by the species. A "procedure" is any use of an animal for experimental, other scientific or educational purposes that may cause pain, suffering, distress or lasting harm equivalent to or higher than that caused by the introduction of a needle in accordance with good veterinary practice.⁹ Article 1(5) excludes six categories regardless of species: non-experimental agricultural practices; non-experimental clinical veterinary practices; veterinary clinical trials required for the marketing authorisation of a veterinary medicinal product; practices undertaken for recognised animal husbandry purposes; practices undertaken for the primary purpose of identifying an animal; and practices not likely to cause pain equivalent to the needle-prick threshold.¹⁰ The comparator in Article 3(1) is therefore not a severity floor within the regulated field but the line between what is regulated at all and what

MevzuatNo=5199&MevzuatTur=1&MevzuatTertip=5 (hereinafter Law No. 5199). Türkiye, Deneysel ve Diğer Bilimsel Amaçlar İçin Kullanılan Hayvanların Refah ve Korunmasına Dair Yönetmelik [Regulation on the Welfare and Protection of Animals Used for Experimental and Other Scientific Purposes], *Resmî Gazete* No. 28141, 13 December 2011, as amended by *Resmî Gazete* No. 28253, 3 April 2012, and No. 30825, 8 July 2019 (hereinafter Welfare Regulation). Türkiye, Hayvan Deneyleri Etik Kurullarının Çalışma Usul ve Esaslarına Dair Yönetmelik [Regulation on the Working Procedures and Principles of Animal Experiment Ethics Committees], *Resmî Gazete* No. 28914, 15 February 2014 (hereinafter Ethics Committee Regulation). It replaced a regulation of the same name published in *Resmî Gazete* No. 26220, 6 July 2006. Türkiye, Bilimsel Amaçlar İçin Kullanılan Sucul Omurgalı Canlıların Refah ve Korunmasına Dair Yönetmelik [Regulation on the Welfare and Protection of Aquatic Vertebrates Used for Scientific Purposes], *Resmî Gazete* No. 30751, 20 April 2019, as amended by *Resmî Gazete* No. 32052, 23 December 2022, and No. 32529, 27 April 2024 (hereinafter Aquatic Regulation).

⁵The consolidated texts were obtained from the official legislation portal (mevzuat.gov.tr) for Law No. 5199, Law No. 5996 and the Aquatic Regulation, and from the Lexpera legal database for the Welfare Regulation and the Ethics Committee Regulation, all accessed 28 September 2026.

⁶B. M. Gölcü and A. Aksoy, "Avrupa Birliği ve Türkiye'de deney hayvanları mevzuatına genel bir bakış [An Overview of Legislation on the Use of Experimental Animals in the European Union and Turkey]," *Türkiye Klinikleri Journal of Laboratory Animals* 1, no. 1 (2017): 56–62, <https://doi.org/10.5336/jlabanim.2016-52876>.

⁷Directive 2010/63/EU, art. 1(2).

⁸Directive 2010/63/EU, art. 1(3). On the inclusion of cephalopods, the first time an entire invertebrate class entered EU laboratory-animal legislation, see G. Fiorito et al., "Guidelines for the Care and Welfare of Cephalopods in Research – A Consensus Based on an Initiative by CephRes, FELASA and the Boyd Group," *Laboratory Animals* 49, no. 2 suppl. (2015): 1–90, <https://doi.org/10.1177/0023677215580006>.

⁹Directive 2010/63/EU, art. 3(1).

¹⁰Directive 2010/63/EU, art. 1(5)(a)–(f).

falls outside the Directive, drawn before any classification under Article 15. Killing animals solely for their organs or tissues is excluded from the definition of procedure but remains regulated, since Article 6 requires killing with minimum pain, in an authorised establishment, by a competent person and by a method listed in Annex IV.¹¹ The result is a two-track structure: killing for tissue escapes project authorisation and severity classification, but not the Directive's welfare and competence requirements. Breeding an animal specifically so that its organs or tissues can be used brings it within scope under Article 1(2), so that the rules on breeding establishments, care and record-keeping apply.

2.2. The Turkish instruments

The Welfare Regulation rests on three statutes: Law No. 5996 on veterinary services, plant health, food and feed, Law No. 5199, and Law No. 5977 on biosafety.¹² It states that it was prepared in parallel with the Directive, and its scope and definitions read as a direct transposition.¹³ Article 2(1)(c) covers live non-human vertebrates, independently feeding larval forms, mammalian foetuses in the last third of development and live cephalopods, matching Article 1(3) of the Directive point for point. Article 2(1)(ç) reproduces Article 1(4) on earlier developmental stages, and Article 2(1)(b) keeps the use within scope even where pain is successfully eliminated by anaesthesia or analgesia. Article 4(s) defines "procedure" in terms nearly identical to Article 3(1) of the Directive, including the needle-prick comparator and the exclusion of killing solely for organs or tissues. The exclusions in Article 2(2) correspond to all six categories in Article 1(5) of the Directive.

Law No. 5199 carries an older and broader definitional layer. It defines an experiment as the use of any animal for experimental or other scientific purposes that will cause pain, suffering, distress or lasting harm, and an experimental animal as any animal used or to be used in an experiment.¹⁴ The statutory definition has no taxonomic limit, and the statute elsewhere defines wild animals so as to include invertebrates; it has no needle-prick comparator; and it refers neither to educational purposes nor to procedures with known or unknown outcome.¹⁵ The Directive-aligned scope therefore exists only at the level of the implementing regulations, and the outer limit of the statutory requirements in Article 9 of Law No. 5199 is set by regulation rather than by statute. The point is narrow but real: the statutory conditions of scientific purpose, use only where no other option exists and ethics-committee authorisation are written for animals without any species boundary. The competent ministry under the statute is the Ministry of Agriculture and Forestry, which since 2018 has held the powers that the two regulations originally divided between the agriculture ministry, responsible for welfare, and the forestry and water ministry, responsible for the ethics committees.¹⁶

Killing for tissue is also partly regulated in the statute. Law No. 5199 prohibits removing a living animal's organs or tissues other than for medical purposes, prohibits non-therapeutic interventions before death has been confirmed, and makes those responsible for killing liable for confirming death and for disposal of the carcass; it bars killing during birth, pregnancy and lactation but expressly excepts medical and scientific grounds.¹⁷ Turkish law thus reproduces the Directive's two-track structure, with killing outside the definition of procedure but still regulated, and anchors part of the second track in primary legislation.

¹¹Directive 2010/63/EU, arts. 3(1) and 6(1)–(3).

¹²Türkiye, Veteriner Hizmetleri, Bitki Sağlığı, Gıda ve Yem Kanunu [Law on Veterinary Services, Plant Health, Food and Feed], Law No. 5996 of 11 June 2010, *Resmî Gazete* No. 27610, 13 June 2010, consolidated text, accessed 28 September 2026, <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5996.pdf> (hereinafter Law No. 5996).

¹³Welfare Regulation, art. 3.

¹⁴Law No. 5199, art. 3(l)–(m).

¹⁵Law No. 5199, art. 3(h).

¹⁶Law No. 5199, art. 3(o).

¹⁷Law No. 5199, arts. 8(2), 13(3)–(4) and 14(1)(e). Breaches carry per-animal fines under art. 28(d), (ı) and (j); for art. 13(4) the fine rises from 2,200 to 5,500 Turkish lira per animal where the breach is committed by an establishment.

One structural feature has no EU counterpart. Since 2019, aquatic vertebrates have been governed by the separate Aquatic Regulation, which covers fish, amphibians, aquatic reptiles other than cetaceans, cyclostomes and cephalopods, and which, since its amendments, runs its own establishment permits and its own central and local ethics committees under the Ministry's Directorate-General of Fisheries and Aquaculture.¹⁸ The Directive's scope is therefore split across two Turkish regulations, and the Turkish home of cephalopod protection is the aquatic instrument. The split matters because fish were the largest single group of animals used in Türkiye in 2020, at 37 per cent.¹⁹ On scope and definitions, the two regimes give the same answer to the same questions; the divergences lie elsewhere.

3. Authorisation and project evaluation

3.1. The Directive

The Directive separates establishment authorisation from project authorisation and assigns both to a competent authority designated by each Member State, which may delegate specific tasks, including project authorisation, to other bodies.²⁰ The substance of evaluation does not travel with that discretion. No project may be carried out without a favourable evaluation, which must weigh the project's objectives, the severity of the procedures and a harm-benefit analysis assessing whether the harm to the animals is justified by the expected outcome, taking ethical considerations into account.²¹ The evaluation must draw on expertise in the scientific field, experimental design, veterinary care and animal husbandry, and must be impartial.²² Applicants must be notified within 40 working days of a complete application, extendable once by up to 15 working days.²³ Authorisation may be granted for at most five years.²⁴ Projects using non-human primates or involving procedures classified as severe must undergo retrospective assessment, and the authority must publish non-technical summaries of authorised projects that are anonymous and state the expected harms and benefits, the number and type of animals, and how the 3Rs are met.²⁵ Separately, each establishment must set up an animal-welfare body, comprising at least the person responsible for the welfare and care of the animals and, in a user establishment, a scientific member, with input from the designated veterinarian. Its tasks are advisory: it guides staff on welfare, acquisition, accommodation, care and the application of the 3Rs, but it does not authorise projects.²⁶ The Directive thus separates three functions: advice inside the establishment, evaluation, which may be delegated, and authorisation, which remains the act of a competent authority.

3.2. Turkish law

Establishment licensing under the Welfare Regulation runs in two stages. The provincial governor's office issues a one-year establishment permit after an on-site inspection by a commission of veterinarians, and the Ministry's Directorate-General then issues a ten-year operating permit after a second inspection.²⁷ Procedures outside a licensed user establishment require a permit from the provincial directorate, valid for the duration of the project and for at

¹⁸Aquatic Regulation, arts. 2 and 4; B. Demirel and G. Uluköy, "Hayvan refahı bakış açısı ile bilimsel çalışmalarda sucül omurgalı canlıların kullanımının mevzuat açısından incelenmesi [A Legislative Evaluation of the Use of Aquatic Vertebrates in Scientific Research from an Animal Welfare Perspective]," *Mediterranean Fisheries and Aquaculture Research* 8, no. 2 (2025): 116–17, <https://doi.org/10.63039/medfar.1818317>.

¹⁹Demirel and Uluköy, "Hayvan refahı bakış açısı," 121.

²⁰Directive 2010/63/EU, art. 59.

²¹Directive 2010/63/EU, arts. 36 and 38(1)–(2).

²²Directive 2010/63/EU, art. 38(3)–(4).

²³Directive 2010/63/EU, art. 41.

²⁴Directive 2010/63/EU, art. 40(3).

²⁵Directive 2010/63/EU, arts. 39(2) and 43(1).

²⁶Directive 2010/63/EU, arts. 26–27.

²⁷Welfare Regulation, arts. 4(k), 11 and 12. The 2019 amendment moved the filing of applications to the provincial directorates and allowed electronic submission.

most three years.²⁸ Inside a licensed establishment, the local ethics committee holds a set of case-by-case powers under the Welfare Regulation. It may allow procedures involving prolonged severe pain, reuse of animals, and the use of listed species from unlicensed sources on scientific grounds, and, with the permission of the relevant ministry, it may make exceptions to the ban on animals taken from the wild; it also applies the replacement and reduction principles, and refinement outside breeding, care and housing.²⁹ Each establishment must also have an animal-welfare unit whose composition and advisory tasks mirror Articles 26 and 27 of the Directive.³⁰

Project authorisation rests on primary legislation. Article 9 of Law No. 5199 allows institutions to conduct animal experiments only through ethics committees established within their own body, and delegates the committees' composition and procedure to a ministerial regulation.³¹ That regulation is the Ethics Committee Regulation, issued, like the Welfare Regulation, in parallel with the Directive.³² HADMEK, the central committee, has 21 members drawn from the ministries, veterinary and medical faculties, the professional associations, TÜBİTAK and one animal-protection organisation. It sets ethical principles, approves and suspends local rulebooks, inspects local committees, hears appeals, sets training rules and must publish annual statistics.³³ A local committee may be established only where the institution has a licensed animal unit, and no experiment may be conducted without both a local committee and an animal-welfare unit. Each local committee has between 5 and 21 members, including a full-time veterinarian with a year's experience and a user certificate, representatives of the units using animals, a citizen with no conflict of interest who does not conduct animal research, and a member of a non-governmental organisation with no ties to the institution. At least one member must hold a doctorate or a medical specialty together with a year's experience of experiments on living animals.³⁴ HADMEK meets quarterly with a two-thirds quorum; local committees meet at least monthly with a two-thirds quorum, decide by majority, and members withdraw from the discussion of their own applications.³⁵

The evaluation itself follows the Directive closely. Projects are assessed against eight criteria: scientific, educational or legal justification; the justification for using animals; humane and environmentally sensitive design; expected benefit; compliance with the 3Rs; severity classification; "the benefit to be gained and the suffering of the animals"; and the lawfulness of killing, anaesthesia, reuse and housing, together with the need for retrospective assessment.³⁶ Reviewers are chosen for expertise in the 3Rs, experimental design, animal experimentation, wildlife or animal care, and the evaluation must be transparent and impartial.³⁷ A committee decides within 40 working days, extendable once by up to 15 working days with reasons; authorisations last at most five years; applications must include a non-technical summary meeting the Directive's content and anonymity requirements; and retrospective assessment is mandatory for projects using primates or severe procedures.³⁸ A small set of activities is exempt from approval: clinical diagnosis and treatment, work on dead animals or tissues, milking, and the collection of faeces or swabs.³⁹ Read directly, the regulation thus answers most of the institutional questions that a comparison of statutes alone would leave open, and nearly all of its answers match the Directive. On these points the Turkish text is closer to the Directive's wording than the transposing laws of several Member States, which reworded or omitted elements of the harm-benefit requirement on transposition. Denmark, for instance, speaks of a

²⁸Welfare Regulation, art. 19(1), as amended in 2019.

²⁹Welfare Regulation, arts. 14(3), 17(2), 22(3), 23(2) and 26(5).

³⁰Welfare Regulation, arts. 8 and 9.

³¹Law No. 5199, art. 9(4)–(5).

³²Ethics Committee Regulation, art. 3.

³³Ethics Committee Regulation, arts. 6 and 7.

³⁴Ethics Committee Regulation, art. 8(1)–(4).

³⁵Ethics Committee Regulation, arts. 6 and 8(8)(a)–(b) and (h).

³⁶Ethics Committee Regulation, art. 15(1).

³⁷Ethics Committee Regulation, art. 15(2)–(3).

³⁸Ethics Committee Regulation, arts. 8(8)(d), (e) and (ğ), 16(1)–(3) and 17(2); compare Directive 2010/63/EU, arts. 39(2), 40(3), 41 and 43.

³⁹Ethics Committee Regulation, art. 8(8)(k).

project's usefulness and omits ethical considerations, Finland drops the outcome element, and national guidance on how to carry out the analysis existed only in France, the Netherlands, Switzerland and the United Kingdom, together with a regional document for Brussels.⁴⁰

3.3. Two divergences

The first divergence is independence. Under the Directive, bodies other than public authorities may perform the competent authority's tasks only if they are free of any conflict of interests.⁴¹ In Türkiye, a committee inside the user institution both evaluates and authorises. Its chair and veterinarian must be employees of that institution, all members are appointed by the rector or head of the institution, and only two of its minimum five members must be free of ties to it.⁴² The requirement of impartiality in Article 15(3) is thus placed inside the institution that wants the project to proceed, and the external checks are limited to HADMEK's review on appeal and its power to suspend a committee.⁴³ The Directive does not require that evaluation be performed by a public authority; it permits delegation, and institutional committees play a role in many Member States. What it does require is that the authorising decision rest with a competent authority, or with a body that is free of conflicts of interest, and that the evaluation be impartial. The Turkish design merges the two stages in the institution that has an interest in the outcome, and it relies on composition rules, rather than on structural separation, to secure impartiality. Two independent members out of a minimum of five is a modest safeguard when the chair, the veterinarian and the other representatives are employees appointed by the same head. The aquatic system is weaker still. Since the 2022 amendment, every permitted aquatic facility must have an aquatic local committee (SOCYEK) of five to eleven members, under a central aquatic committee (SOCMEK) recomposed in 2024. The chair and the facility manager must be employees, the rector or head appoints all members, and only one member, drawn from an animal-related non-governmental organisation, must have no ties to the institution. SOCMEK, chaired by a senior official of the Ministry, approves the local rulebooks, inspects the local committees and must collect and publish annual statistics on aquatic animal use.⁴⁴

The second divergence is drift between the regulation and the rulebooks that each committee writes for itself and HADMEK must approve.⁴⁵ Three rulebooks were read for this study. That of the İzmir Institute of Technology reproduces the regulation, including the 40-working-day period and the benefit-versus-suffering criterion. That of İstanbul Medipol University provides for decisions "at the latest within two meetings", sets a simple-majority quorum where the regulation requires two-thirds, and omits the benefit-versus-suffering test from its working principles. The rulebook of İstanbul University-Cerrahpaşa, approved by HADMEK in 2021, was available only as a scanned copy and was not analysed in detail.⁴⁶ A sample of three proves nothing about the system as a whole, but it shows that the regulation's standards do not reach every committee's own rules unchanged, even after central approval.

⁴⁰D. Hajosi and H. Grimm, "Mission Impossible Accomplished? A European Cross-National Comparative Study on the Integration of the Harm-Benefit Analysis into Law and Policy Documents," *PLOS ONE* 19, no. 2 (2024): e0297375, 6–9, 19–20, <https://doi.org/10.1371/journal.pone.0297375>.

⁴¹Directive 2010/63/EU, art. 59(1).

⁴²Ethics Committee Regulation, art. 8(2)–(4).

⁴³Ethics Committee Regulation, arts. 7(1)(f) and 21(2).

⁴⁴Aquatic Regulation, arts. 31/A–31/C, added by *Resmî Gazete* No. 32052, 23 December 2022; art. 31/A(1) as amended by *Resmî Gazete* No. 32529, 27 April 2024; on the practical operation of the system, including five-year operating permits, project approvals of up to three years and recurring uncertainty over which activities, such as stripping eggs and sperm from broodstock, count as procedures, see Demirel and Uluköy, "Hayvan refahı bakış açısı," 116–19 and 122.

⁴⁵Ethics Committee Regulation, arts. 7(1)(b) and 8(8)(c).

⁴⁶İzmir Yüksek Teknoloji Enstitüsü, *İYTE Hayvan Deneyleri Yerel Etik Kurulu Yönergesi* [Local Animal-Experiment Ethics Committee Rulebook] (İzmir: İYTE, 2014); İstanbul Medipol Üniversitesi, *Hayvan Deneyleri Yerel Etik Kurulu Yönergesi* [Local Animal-Experiment Ethics Committee Rulebook] (İstanbul: Medipol Üniversitesi, n.d.); İstanbul Üniversitesi-Cerrahpaşa, *Hayvan Deneyleri Yerel Etik Kurulu (İÜC-HADYEK) Yönergesi* [Local Animal-Experiment Ethics Committee Rulebook] (İstanbul: İÜC, 2021).

Law No. 5199 also states substantive thresholds directly: animals may not be used in non-scientific experiments, must be protected, cared for and housed appropriately, and may be used only "in the absence of any other option".⁴⁷ One statutory control has no counterpart in the Directive: the import and export of experimental animals requires a ministerial permit.⁴⁸

4. Severity classification and the 3Rs

Textually, the two regimes are close to identical. Article 4 of the Directive states replacement, reduction and refinement as binding obligations; Article 26 of the Welfare Regulation mirrors the same structure, and Law No. 5199 independently states replacement and a general refinement-type duty at statutory level.⁴⁹ Reduction has no statutory anchor. The Turkish placement of replacement in primary legislation, with the formula that animals may be used only in the absence of any other option, is in one respect stronger than the Directive's own architecture, in which the principle sits in secondary Union law transposed by national implementing rules. Severity classification converges as well: Article 22 of the Welfare Regulation requires the same case-by-case, four-tier classification as Article 15 of the Directive, using criteria in Annex 8 that reproduce Annex VIII of the Directive, and Annex 3 of the Aquatic Regulation does the same.⁵⁰ The rules on choice of methods and on anaesthesia track Articles 13 and 14 of the Directive with comparable closeness.

One divergence survives, and it is institutional rather than definitional. Article 15(2) of the Directive prohibits procedures involving severe pain, suffering or distress that is likely to be long-lasting and cannot be ameliorated. An exception requires a Member State to invoke the safeguard clause, adopting a provisional measure and informing the Commission and the other Member States immediately.⁵¹ The Welfare Regulation permits the same exception but grants the power to any local ethics committee, case by case, on scientific grounds, and the Aquatic Regulation repeats the rule; neither was changed by later amendments.⁵² The Ethics Committee Regulation adds two safeguards: the committee must weigh the benefit against the animals' suffering before allowing severe, prolonged procedures, and such projects must undergo retrospective assessment.⁵³ What the Turkish texts do not contain is any notification to a national authority. A decision that the Union treats as a state measure requiring international visibility is, in Türkiye, a local committee's discretionary call reviewed after the fact.

Beyond that divergence, the texts do not distinguish the regimes: both bind on paper. What binding means is itself contested. Grimm and Dusseldorp argue that the 3Rs are principles in the norm-theoretical sense, commands to realise their content to the highest degree possible, measured against the Directive's stated goal of full replacement.⁵⁴ Neither regime's text says which reading its review bodies should apply. Whether the classification and 3Rs obligations are applied with comparable rigour is a question of enforcement capacity, to which section 6 returns.

5. Personnel competence

Article 23 of the Directive requires adequate education and training before any of four functions is performed: carrying out procedures, designing procedures and projects, taking care of animals and killing animals. Designers must also have training in a relevant scientific discipline and species-specific knowledge, and staff in the other functions work under

⁴⁷Law No. 5199, art. 9(1)–(3).

⁴⁸Law No. 5199, art. 10(7).

⁴⁹Law No. 5199, art. 9(2)–(3); Welfare Regulation, art. 26.

⁵⁰Welfare Regulation, art. 22(1) and Annex 8; Aquatic Regulation, art. 20(1) and Annex 3.

⁵¹Directive 2010/63/EU, arts. 15(2) and 55(3)–(4).

⁵²Welfare Regulation, art. 22(2)–(3); Aquatic Regulation, art. 20(2)–(3).

⁵³Ethics Committee Regulation, arts. 10(m) and 17(2).

⁵⁴H. Grimm and M. Dusseldorp, "The (Re)turn of the 3Rs: An Inquiry into the Normative Nature of Russell and Burch's Principles of Humane Experimental Technique," *Laboratory Animals* 59, no. 5 (2025): 557–60, <https://doi.org/10.1177/00236772251326352>.

supervision until they have demonstrated competence.⁵⁵ Article 32 of the Welfare Regulation reproduces this four-function structure together with the designer-specific and supervision rules, and its Annex 10 lists eleven minimum training elements that translate Annex V of the Directive one for one; Annex 4 of the Aquatic Regulation repeats the list.⁵⁶ The curriculum therefore matches the Directive. What differs is who must complete it and who certifies it.

The Turkish regime departs from Article 23 in two opposite directions. It is looser on training. From 1 January 2013, a facility manager must have at least two years' experience with laboratory animals, and only a manager who lacks that experience, and persons performing procedures, must complete the Annex 10 training.⁵⁷ Experience can thus stand in for training, which Article 23 does not allow. The Aquatic Regulation goes further: two years' experience or Annex 4 training qualifies a manager, and university managers and researchers holding a doctorate are exempt from training altogether.⁵⁸

It is stricter on profession. The Welfare Regulation defines the facility manager as a veterinarian, and animals are killed by a veterinarian or under a veterinarian's supervision.⁵⁹ Experiments on live animals are carried out under the supervision of the responsible veterinarian, research teams working with farm animals must include a veterinarian, and every local committee must include a full-time veterinarian.⁶⁰ The Directive asks only that killing be done by a competent person and allows a suitably qualified expert in place of a designated veterinarian, an option some veterinary commentators would remove.⁶¹ The Aquatic Regulation breaks the pattern by admitting aquaculture and fisheries engineers, biologists, zoologists and hydrobiologists as managers.⁶²

Certification is decentralised. Anyone handling animals in research must hold a user certificate, a committee may not approve a project whose staff lack one, each local committee runs its own courses, requiring 80 per cent attendance and a mark of 70 out of 100, and the certificate is signed by the committee chair and the head of the institution; HADMEK sets course content and judges the equivalence of certificates issued elsewhere.⁶³ Course content was first fixed by a 2007 ministerial circular, which defined three categories: A1 for animal caretakers, with 31 hours of instruction, and A2 for technicians and B for researchers, each with 40 hours of theory and 40 of practice. HADMEK acknowledged in 2017 that the circular had not been updated to the Directive and began to revise it; its replacement, Circular 2019/3, is in force, and the central committee's list of certificate courses for 2026 is issued under it. The text of Circular 2019/3 itself was not available for this study.⁶⁴ Because competence is certified institution by institution, no systematic mapping onto the function categories of the Federation of European Laboratory Animal Science Associations (FELASA) exists, and portability of Turkish certificates abroad is not automatic.⁶⁵

Law No. 5199 anchors this regime. It delegates staff qualifications to regulation, doubles administrative fines where the offender is a veterinarian, veterinary health technician or animal carer, and provides without qualification that medical and surgical interventions on animals are

⁵⁵Directive 2010/63/EU, art. 23(1)–(2).

⁵⁶Welfare Regulation, art. 32(2)–(4) and Annex 10; Aquatic Regulation, Annex 4.

⁵⁷Welfare Regulation, art. 32(4).

⁵⁸Aquatic Regulation, art. 29(2).

⁵⁹Welfare Regulation, arts. 4(s) and 30(1)(b).

⁶⁰Ethics Committee Regulation, arts. 8(2)(a), 10(g) and 18(c).

⁶¹Directive 2010/63/EU, arts. 6(2) and 25; Ł. Kiraga and A. Dzikowski, "Ethical Concerns of the Veterinarian in Relation to Experimental Animals and in Vivo Research," *Animals* 13, no. 15 (2023): 2476, 8, <https://doi.org/10.3390/ani13152476>.

⁶²Demirel and Uluköy, "Hayvan refahı bakış açısı," 118.

⁶³Ethics Committee Regulation, art. 18(a)–(d), (f) and (ğ)–(m).

⁶⁴Hayvan Deneyleri Merkezi Etik Kurulu [HADMEK], 2015–2016 yılları Hayvan Deneyleri Merkezi Etik Kurulu faaliyet raporu [Activity Report 2015–2016] (Ankara: T.C. Orman ve Su İşleri Bakanlığı, 2017); HADMEK, 2026 yılı deney hayvanı kullanım sertifikası eğitim programları listesi [List of 2026 Laboratory-Animal User Certificate Programmes], 25 September 2026, accessed 28 September 2026, <https://hadmek.tarimorman.gov.tr/Duyuru/Detay/16896>.

⁶⁵FELASA, "Education & Training."

carried out only by veterinarians.⁶⁶ None of the regulations says whether non-veterinarian researchers may perform surgical procedures in an approved project under a veterinarian's supervision. If Article 7(1) were read to exclude them, it would amount to a divergence from Article 23 in the opposite direction: stricter rather than looser.

6. Inspection, sanctions and reporting

6.1. Inspection

Internationally, inspection systems vary widely and are not always regarded as effectively implemented.⁶⁷ Article 34 of the Directive requires risk-based inspection of at least one third of users each year, annual inspection of primate facilities and an appropriate proportion of unannounced inspections. The Welfare Regulation copies this and goes further on surprise visits: inspections follow a risk analysis using the Directive's criteria, must reach at least one third of establishments a year and primate facilities annually, and at least half must be unannounced.⁶⁸ The Ethics Committee Regulation subjects all its provisions to ministerial inspection on HADMEK's opinion and gives HADMEK its own power to inspect committees and laboratories, and Law No. 5199 vests inspection authority in the Ministry, delegable to the provincial governor.⁶⁹

Two gaps survive. The Aquatic Regulation keeps the risk analysis and the 50 per cent unannounced rule but omits the one-third annual floor, and its 2022 and 2024 amendments left that article unchanged.⁷⁰ The floor is therefore absent for the largest group of animals used. Nor is there any public record of whether the floor is met elsewhere. No count of ministerial inspections under Article 37 of the Welfare Regulation was found. The only published figures are HADMEK's own: 33 central-committee inspections between 2008 and 2015, of which 22 took place in 2009 and none in 2012 or 2014.⁷¹ These are a track separate from the Ministry's duty and are not evidence that it is breached, but they are the only observable data. By contrast, the Directive exposes national inspection systems to Commission controls and requires Member States to report on their operation, so that the frequency and outcome of inspections are at least visible at Union level.⁷² No equivalent exists in Turkish law, which contains no obligation to publish inspection counts or findings. Against roughly ninety active local committees, they show occasional central scrutiny rather than a cycle.

6.2. Sanctions

Article 60 of the Directive requires penalties that are effective, proportionate and dissuasive, a qualitative standard that Member States implement and notify to the Commission and that is deliberately left open to scale with the violation and the violator. Internationally, sanctions range from fines through imprisonment to the withdrawal of authorisation or accreditation.⁷³ Turkish sanctions run on two statutory tracks, both with fixed schedules. Breaches of the Welfare Regulation are sanctioned under Article 36 of Law No. 5996, alongside suspension of activity, permit revocation and confiscation.⁷⁴ The base amounts are small: 100 Turkish lira per animal for failing to meet animals' welfare, housing or feeding needs; 5,000 lira and a ban for operating without approval; 1,000 lira for failing to keep records; and 50 lira per

⁶⁶Law No. 5199, arts. 7(1), 9(6) and 28.

⁶⁷K. Bayne, T. H. Morris and M. P. France, "Legislation and Oversight of the Conduct of Research Using Animals: A Global Overview," in *The UFAW Handbook on the Care and Management of Laboratory and Other Research Animals*, 8th ed., ed. R. Hubrecht and J. Kirkwood (Chichester: Wiley-Blackwell, 2010), 111.

⁶⁸Welfare Regulation, art. 37.

⁶⁹Ethics Committee Regulation, arts. 7(1)(c)–(d) and 20; Law No. 5199, arts. 17(1) and 25.

⁷⁰Aquatic Regulation, art. 32.

⁷¹HADMEK, *2010–2014 yılları deney hayvanı kullanımı faaliyet raporu* [Activity Report on the Use of Experimental Animals 2010–2014] (Ankara: T.C. Orman ve Su İşleri Bakanlığı, 2015).

⁷²Directive 2010/63/EU, arts. 35 and 54(1).

⁷³Bayne, Morris and France, "Legislation and Oversight," 111.

⁷⁴Welfare Regulation, art. 39; Aquatic Regulation, art. 34.

animal and 200 lira per establishment for failing to identify or register animals.⁷⁵ Breaches of the Ethics Committee Regulation and unauthorised experiments fall under Article 28 of Law No. 5199: 1,200 lira per animal for general non-compliance and 4,500 lira per animal for experiments without authorisation, on the 2021 base.⁷⁶ HADMEK may suspend a non-compliant local committee for one month, then six months, and then cancel its rulebook.⁷⁷

Two features of these schedules matter more than the amounts. First, they scale along the wrong axis for deterrence. The fine is multiplied by the number of animals, doubled for veterinary and care staff, and increased on repetition, and a ban on keeping animals may follow.⁷⁸ None of these escalators tracks the offender's resources, so the same schedule applies to a small academic laboratory and to a facility running internationally sponsored preclinical work. Second, the sanctions are exclusively administrative. The criminal offences in Article 28/A of Law No. 5199 cover killing endangered animals, killing or torturing domestic animals, sexual assault on animals and animal fighting; none is framed around experimental use, and prosecution requires a written application by the Ministry's provincial directorate. Violations involving experimental animals therefore remain administrative infractions rather than crimes. One interpretive edge remains open, since Article 28/A(2) contains no exception for scientific use, but the statutory definition of a domestic animal as one cultured and trained by humans makes its application to purpose-bred laboratory animals unlikely.⁷⁹ A further gap is visible only in the statute: the import and export permit in Article 10(7) has no matching fine, since Article 28(f) sanctions other paragraphs of Article 10 but not the seventh.

6.3. Reporting and transparency

Since Regulation (EU) 2019/1010, non-technical project summaries are published in an open EU database within six months of authorisation, and statistics are submitted electronically and published annually.⁸⁰ Türkiye has most of this on paper. Establishments must file annual records with the provincial directorate, local committees report to HADMEK, and HADMEK must collect and publish annual national statistics.⁸¹ Non-technical summaries are required in every application, but they are published only if the Ministry creates a database, and no such database was found.⁸² HADMEK has published its statistics in multi-year reports; an earlier, unpublished central-committee report counted 148,957 animals in 2008.⁸³ The reports cover 2010–2014 (published in 2015), 2015–2016 (2017), 2017 (2018) and 2018–2020 (2021). The series ends in 2020, although the 2010–2014 report promised annual reports from 2015 onwards. Even the Union's figures are incomplete: around 8 million animals falling under the Directive, alongside the roughly 10 million counted each year, were not included in the headline count.⁸⁴

⁷⁵Law No. 5996, art. 36(1)(e), (g) and (i). Only the transport-document fines in art. 36(1)(f) were raised recently, to 26,360 lira per vehicle for most species, by Law No. 7584 of 2026. Administrative fines are revalued annually, so current amounts exceed these base figures; the current amounts were not verified for this study.

⁷⁶Law No. 5199, art. 28(e); Ethics Committee Regulation, art. 21(1). The regulation cites art. 28(1)(f); in the current text the art. 9 fines are in art. 28(e), a cross-reference apparently left behind when the paragraphs were re-lettered in 2021.

⁷⁷Ethics Committee Regulation, art. 21(2).

⁷⁸Law No. 5199, arts. 24, 28 and 30.

⁷⁹Law No. 5199, arts. 3(e) and 28/A.

⁸⁰European Union, Regulation (EU) 2019/1010 of the European Parliament and of the Council of 5 June 2019 on the alignment of reporting obligations in the field of legislation related to the environment, *Official Journal of the European Union* L 170, 25 June 2019, 115–27; F. Busquet, "New European Union Statistics on Laboratory Animal Use – What Really Counts!," *ALTEx* 37, no. 2 (2020): 167–69, <https://doi.org/10.14573/altex.2003241>.

⁸¹Welfare Regulation, art. 34(2); Ethics Committee Regulation, arts. 7(1)(h) and 9(1)(g).

⁸²Ethics Committee Regulation, art. 16(4).

⁸³S. İzmirli et al., "Introducing Ethical Evaluation of the Use of Animals in Experiments in the Near East," *Alternatives to Laboratory Animals* 38, no. 4 (2010): 331–36, <https://doi.org/10.1177/026119291003800410>.

⁸⁴Busquet, "New European Union Statistics," 167.

Table 2. Animals used for scientific purposes in Türkiye, 2010–2020, as reported by HADMEK (figures read from the reports' charts)⁸⁵

Year	Animals used	Of which fish
2010	270,307	55,275
2011	201,606	50,035
2012	173,152	54,039
2013	167,634	30,226
2014	213,366	45,894
2015	386,745	216,418
2016	451,914	249,213
2017	265,109	132,498
2018	236,297	68,057
2019	213,641	76,377
2020	209,212	77,435

The series is volatile. Total use fell from 270,307 animals in 2010 to 167,634 in 2013, rose to a peak of 451,914 in 2016 and returned to 209,212 in 2020. The share of fish moved with it, from about 20 per cent in 2010 to 55 per cent in 2016 and 37 per cent in 2020. Swings of this size in a national series are more plausibly explained by changes in recording practice than by changes in research activity, which is consistent with the central committee's own warnings discussed below.

The series shows two features that matter more than its trend. Fish drive the swings, and the 2015–2016 peak is almost entirely fish. HADMEK itself warned that the data are not comparable with the Union's, because veterinary clinical research had been counted as experimental use and local committees needed training to complete the statistical tables; in 2011, 14.1 per cent of Turkish use was recorded under disease diagnosis, against 1.6 per cent in the Union.⁸⁶

At project level, the texts lean against transparency. The Welfare Regulation forbids any public statement about procedures without the consent of the person responsible for them, leaving disclosure to the Ministry, and the Aquatic Regulation repeats the rule; the Ethics Committee Regulation makes committee correspondence confidential while also requiring that the public be given objective information about projects.⁸⁷ Reporting runs inward to the state rather than outward to the public.

7. Assessment: six divergences and an observability gap

Read provision by provision, Turkish law is a faithful transposition of Directive 2010/63/EU, not only of its substantive standards but also of much of its institutional machinery: the harm-benefit test with named reviewer expertise, the decision period, the five-year authorisation limit, non-technical summaries, retrospective assessment and a risk-based inspection floor with a higher share of unannounced visits than the Directive requires. Six divergences survive, each narrower than a textual comparison of the statute alone would suggest (Table 1).

⁸⁵HADMEK, 2010–2014 faaliyet raporu; HADMEK, 2015–2016 faaliyet raporu; HADMEK, 2017 yılı hayvan deneyleri istatistik raporu [Statistical Report on Animal Experiments 2017] (Ankara: T.C. Orman ve Su İşleri Bakanlığı, 2018); HADMEK, 2018–2020 yılları Hayvan Deneyleri Merkezi Etik Kurulu faaliyet raporu [Activity Report 2018–2020] (Ankara: T.C. Tarım ve Orman Bakanlığı, 2021).

⁸⁶HADMEK, 2010–2014 faaliyet raporu.

⁸⁷Welfare Regulation, art. 34(1); Aquatic Regulation, art. 30; Ethics Committee Regulation, art. 22(1) and (3).

Table 1. Divergences between Directive 2010/63/EU and Turkish law that survive a provision-level comparison

Issue	Directive 2010/63/EU	Turkish law	Assessment
Independence of project evaluation and authorisation	Competent authority; delegated bodies must be free of any conflict of interests (arts. 38(4), 59(1)).	In-house committee evaluates and authorises; chair and veterinarian are employees; all members appointed by the institution; two independent members (one on aquatic committees).	Impartiality rests on lay members and central review on appeal.
Severe, long-lasting pain	Prohibited unless the Member State adopts a provisional measure notified to the Commission (arts. 15(2), 55(3)).	Any local committee may allow it case by case; retrospective assessment required; no notification.	A state measure in the Union is a local decision in Türkiye.
Aquatic inspection and committees	At least one third of users inspected each year (art. 34).	Aquatic Regulation omits the one-third floor; one independent committee member.	Weaker rules for the largest group of animals used.
Transparency	Summaries published in an EU database; annual statistics (arts. 43, 54).	Summaries published only if the Ministry creates a database; confidentiality provisions; statistics end in 2020.	No project-level transparency in practice.
Personnel	Competence through training (art. 23, Annex V); competent person or expert (arts. 6, 25).	Experience can replace training for managers; doctorate exemption for aquatic work; veterinarian-centred roles.	Looser on training, stricter on profession.
Sanctions	Effective, proportionate and dissuasive (art. 60).	Fixed per-animal schedules under Laws No. 5996 and 5199; administrative only.	Sanctions do not scale with the offender's resources.

These findings qualify the earlier criticism that the Turkish regulations were translated almost word for word from the Directive without adaptation.⁸⁸ The close translation is confirmed, but the provision-level comparison shows that it has been selective. Where the Turkish instruments adapt the Directive, they do so in ways that keep decisions local and inside institutions, as with project authorisation and the severe-pain exception, or that reflect the domestic weight of the veterinary profession, as with the personnel rules. The adaptation that is missing is the one that would make the system legible from outside: publication of summaries, of inspection data and of up-to-date statistics.

The first divergence, independence, is the most structural. It does not follow that Turkish committees are partial, and no evidence was found that they are; it follows only that the design gives no external assurance that they are not. The second, the severe-pain exception, concerns a small number of projects but a large principle: the Directive treats the authorisation of long-lasting, unrelievable suffering as a matter of state responsibility and international accountability, while Turkish law treats it as a matter of local scientific judgement subject to retrospective review.

The third divergence is the product of the 2019 carve-out of aquatic vertebrates. It is paradoxical that the rules are weakest, on inspection frequency, committee independence and training exemptions, for the group of animals that has recently accounted for the largest share of use. The fourth, transparency, is partly a matter of omission, since the summary database has not been created, and partly a matter of design, since three provisions restrict disclosure and none requires publication of inspection results.

⁸⁸Gölcü and Aksoy, "Avrupa Birliği ve Türkiye'de," 56–62.

The fifth and sixth divergences pull in different directions. The personnel rules are at once looser than the Directive, because experience can replace training, and stricter, because professional roles are reserved to veterinarians; the net effect on competence cannot be assessed from the texts. The sanctions are fixed, low and exclusively administrative. Neither divergence would, on its own, prevent a well-run facility from meeting Union standards, but both weaken the capacity of the state to compel a poorly run one to do so.

The larger divergence is observational. The Directive's chain of obligations is not only written down but reported on, through open project summaries, annual statistics, five-yearly implementation reports and Commission controls of national inspection systems.⁸⁹ Türkiye's chain is written in nearly the same words, but its performance leaves few public traces: no inspection counts under Article 37, no summary database, national statistics that stop in 2020 and that their own publisher warns against comparing, and local rulebooks that do not always carry the regulation's standards. The law is largely adequate; its application cannot be checked from outside. Observability, in this sense, is itself a legal attribute of a regulatory regime. It depends on duties to record, to report and to publish, and on the existence of bodies with the mandate to aggregate and disclose what is recorded. The Directive builds those duties into its chain of obligations; the Turkish instruments reproduce most of the recording duties but few of the publication duties, and the one publication duty that matters most for projects is made conditional on an administrative act that has not occurred. This has a practical consequence for international research cooperation. Assurance about a Turkish facility continues to come from facility-specific evidence, such as the local committee's rulebook and minutes, inspection reports held by the facility, accreditation status and sponsor audits, rather than from the public regulatory record. International accreditation offers only a partial substitute: AAALAC International lists a single accredited Turkish institution.⁹⁰ Fixed fine schedules that do not scale with the offender's resources are, for an internationally linked facility, a weaker deterrent than the loss of accreditation or of a sponsor's contract.

8. Conclusions

Turkish law on animals used for scientific purposes is not thin. Its regulations reproduce Directive 2010/63/EU with unusual fidelity, and at some points, notably the harm-benefit wording and the share of unannounced inspections, they are closer to the Directive than the laws of several Member States. The divergences that remain concern independence, the severe-pain exception, aquatic inspection and committee composition, transparency, the balance between professional status and training, and the structure of sanctions. Each is narrower than the general claim that the Turkish system lacks the Directive's institutional rules.

The decisive gap lies in practice, and it is also a research gap. No study was found that examines how Turkish local committees work: approval rates, turnaround against the 40-day period, use of the severe-pain exception, or the number of ministerial inspections. Descriptive and legal accounts exist, but the empirical layer is unstudied.⁹¹ A tractable first step would be a systematic comparison of published committee rulebooks with the 2014 regulation, which the three rulebooks read here suggest would reveal further drift. A second would be a request, under the right to information, for the number of ministerial inspections carried out under Article 37 of the Welfare Regulation and Article 32 of the Aquatic Regulation. A third would be a survey of committee practice on the harm-benefit test, the 40-day period and the severe-pain exception. Two interpretive questions under Law No. 5199 also await practice or case law: whether the veterinarian-only rule for surgical interventions limits who may operate in an approved project, and whether any experimental conduct could fall within the criminal offences of Article 28/A.

⁸⁹Directive 2010/63/EU, arts. 35, 43 and 54.

⁹⁰AAALAC International, "Directory of Accredited Organizations."

⁹¹See Gölcü and Aksoy, "Avrupa Birliği ve Türkiye'de"; Ö. Uludağ, "Hayvan deneyi çalışmalarında etik kuralların tarihçesi ve önemi [History and Importance of Ethical Rules in Animal Experiment Studies]," *Adıyaman Üniversitesi Sağlık Bilimleri Dergisi* 5, no. 1 (2019): 1401–13, <https://doi.org/10.30569/adiyamansaglik.482098>; Demirel and Uluköy, "Hayvan refahı bakış açısı."

The direction of travel is uncertain. Alignment has run through domestic regulation modelled on Union law, not through treaty obligation: Türkiye signed the Council of Europe Convention for the Protection of Vertebrate Animals used for Experimental and Other Scientific Purposes (ETS No. 123) in 1986, but the Convention has not entered into force for it.⁹² The framework has not been frozen: the Aquatic Regulation and its amendments, and the 2019 training circular, show continued revision after accession talks stalled. Whether Turkish law will follow later changes on the Union side, such as the 2024 update of the Directive's Annex IV on killing methods, remains an open question. For accession policy, the lesson is that screening of alignment in Chapter 12 cannot stop at the text: a regime that matches the Directive on paper may still lack the reporting infrastructure through which the Union itself verifies implementation. For the transparency that the Directive now demands, the missing element in Türkiye is less a new rule than the publication of what existing rules already require: summaries, statistics and inspection records.

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⁹²D. Akben, "Alman hukukunda ve Türk hukukunda hayvanların korunmasının karşılaştırılması [A Comparison of Animal Protection in German and Turkish Law]," *Dokuz Eylül Üniversitesi Hukuk Fakültesi Dergisi* 25, no. 2 (2023): 1047–1102, <https://doi.org/10.33717/deuhfd.1375617>.

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